Deep Learning for Five-Class Alzheimer's Disease Classification Using Structural MRI: A Comparative Study of CNN Architectures

CAP5626 — Introduction to Artificial Intelligence

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Abstract

Alzheimer's disease (AD) is a progressive neurodegenerative disorder for which early and accurate diagnosis is critical. This project presents a complete deep learning pipeline for five-class classification of structural MRI brain images into AD, Cognitively Normal (CN), Early Mild Cognitive Impairment (EMCI), Late Mild Cognitive Impairment (LMCI), and Mild Cognitive Impairment (MCI). Five CNN architectures are systematically designed, trained, and compared: a custom CNN from scratch, VGG-16, ResNet-50, DenseNet121, and EfficientNet-B0 — all discussed in Dr. Theran's CNN Architectures lecture [10]. The pipeline incorporates five preprocessing steps validated through empirical four-pipeline comparison, seven

data augmentation techniques covering all handout examples, strict subject-level data split (489 unique patients, zero overlap verified), class-imbalance correction via weighted random sampling, and three overfitting mitigation strategies. Without augmentation, the model collapsed to predicting a single class (macro F1 = 0.093). DenseNet121 achieved the best performance across all primary metrics: macro ROC-AUC = 0.6535, macro F1 = 0.3201, PR-AUC = 0.4002, and LMCI recall = 0.5455. These results directly validate Dr. Theran's CNN Architectures lecture (slide 8), which identifies DenseNet as best suited for "smaller datasets, feature reuse-heavy problems" [10].

Contents

1 Introduction **6**

2 Related Work **6**

2.1 Reference Paper 6

2.2 CNN Architecture Families 7

2.3 Deep Learning in Medical Imaging 7

2.4 Transfer Learning 7

2.5 Class Imbalance 7

2.6 Evaluation Metrics 7

3 Dataset Description **8**

3.1 Source and Format 8

3.2 Class Distribution 8

3.3 Train, Validation, and Test Splits 10

3.4 Data Leakage Prevention 10

4 Preprocessing Pipeline **10**

4.1 Preprocessing Study 10

4.2 Comparative Pipeline Results 11

4.3 Selected Pipeline and Scientific Justification 11

4.4 Steps Not Empirically Testable 13

5 Data Augmentation **13**

5.1 Why Augmentation Is Necessary 13

5.2 Seven Augmentation Techniques 13

5.3 With vs. Without Augmentation 14

6 Model Architectures **15**

6.1 Architecture 1 — Custom CNN from Scratch 15

6.2 Architecture 2 — VGG-16 Transfer Learning 16

6.3 Architecture 3 — ResNet-50 Transfer Learning 17
6.4 Architecture 4 — DenseNet121 Transfer Learning 17
6.5 Architecture 5 — EfficientNet-B0 Transfer Learning 17

7 Training Strategy **18**

8 Overfitting Analysis **18**
8.1 Strategy 1 — Dropout and Batch Normalization 18
8.2 Strategy 2 — Early Stopping 18
8.3 Strategy 3 — Learning Rate Scheduling 19
8.4 Overfitting per Architecture 19
8.5 Strategy 7 — Class Weighting via Sampling 20
8.6 Effectiveness Ranking of Mitigation Strategies 20

9 Results and Evaluation **21**
9.1 Evaluation Metrics 21
9.2 Overall Results 22
9.3 Per-Class Analysis — DenseNet121 (Best Model) 22
9.4 Per-Class Analysis — VGG-16 22
9.5 Per-Class F1 Comparison 23
9.6 Confusion Matrices 23

10 Discussion **24**
10.1 Scientific Architecture Comparison 24
10.2 Which Classes Are Easier or Harder to Distinguish 24
10.3 Complete Scientific Comparison 25
10.4 Why DenseNet121 Achieves Best Performance 26
10.5 Why Accuracy Is Misleading 26
10.6 VGG-16 Overfitting Analysis 27
10.7 Impact of Patient-Level Split 27
10.8 Strengths, Weaknesses, and Limitations 27

11 Conclusion and Future Work **28**

1 Introduction

Alzheimer's disease is the most prevalent form of dementia, accounting for 60–80% of all dementia cases worldwide [1]. The disease progresses through distinct clinical stages — from early mild cognitive impairment through late MCI to confirmed Alzheimer's — making early-stage classification clinically valuable for timely intervention. Structural MRI provides a non-invasive window into neurodegeneration, capturing brain atrophy patterns associated with AD progression, particularly in the hippocampus and entorhinal cortex.

Deep learning, particularly CNNs, has emerged as a powerful tool for automated medical image analysis [6]. Transfer learning from large natural image datasets (e.g., ImageNet) further extends CNN utility to medical domains where labeled data is scarce [7].

This project builds and evaluates five CNN-based classification pipelines for five diagnostic categories: AD, CN, EMCI, LMCI, and MCI. Central scientific challenges include: (1) small dataset size (1,101 training images), (2) severe class imbalance (LMCI: only 5.5% of training data), (3) fine-grained visual similarity between adjacent disease stages, and (4) ensuring strict subject-level data separation. All five architectures are drawn from Dr. Theran's CNN Architectures lecture [10].

2 Related Work

2.1 Reference Paper

Lim et al. [1] proposed a deep learning framework for AD classification using structural MRI from the ADNI dataset, comparing a custom CNN against VGG-16 and ResNet-50. They identified overfitting as the dominant challenge, persisting even with dropout, batch normalization, augmentation, and early stopping.

2.2 CNN Architecture Families

He et al. [2] introduced ResNet with residual connections $\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x}$, enabling very deep networks by mitigating vanishing gradients. Huang et al. [3] proposed DenseNet, where each layer receives feature maps from all preceding layers via concatenation — Dr. Theran's lecture identifies DenseNet as best for small datasets [10]. Tan and Le [5] introduced EfficientNet, which jointly scales depth, width, and resolution using a compound coefficient. Simonyan and Zisserman [4] proposed VGG, explicitly listed in the project handout §3.4 and compared in Lim et al. [1].

2.3 Deep Learning in Medical Imaging

Litjens et al. [6] surveyed deep learning across medical imaging modalities, establishing CNNs as dominant and documenting that transfer learning consistently outperforms random initialization on small medical datasets.

2.4 Transfer Learning

Tajbakhsh et al. [7] demonstrated that ImageNet pretrained CNNs consistently outperform randomly initialized networks on medical imaging, even across domain shift.

2.5 Class Imbalance

Johnson and Khoshgoftaar [8] found weighted random sampling more robust than loss weighting for severe imbalance, recommending combining oversampling with standard cross-entropy loss.

2.6 Evaluation Metrics

As detailed in Dr. Theran's evaluation metrics lecture [9], accuracy is misleading under class imbalance. ROC-AUC provides threshold-independent ranking quality and PR-AUC is more informative when positives are rare.

3 Dataset Description

3.1 Source and Format

The dataset was provided by Dr. Carlos Theran (FAMU, CAP5626) and is derived from the Alzheimer's Disease Neuroimaging Initiative (ADNI) [1]. Images are 2D axial MRI slices in JPEG format organized into five class folders.

3.2 Class Distribution

Table 1: Dataset class distribution.

Class	**Train**	**Test**	**Train %**
AD	145	26	13.2%
CN	493	87	44.8%
EMCI	204	36	18.5%
LMCI	61	11	5.5%
MCI	198	35	18.0%
Total	**1,101**	**195**	100%

The dataset is severely imbalanced: CN represents 44.8% of training samples while LMCI comprises only 5.5% — an 8.1:1 ratio that directly motivates the class-imbalance correction strategy (Section 7).

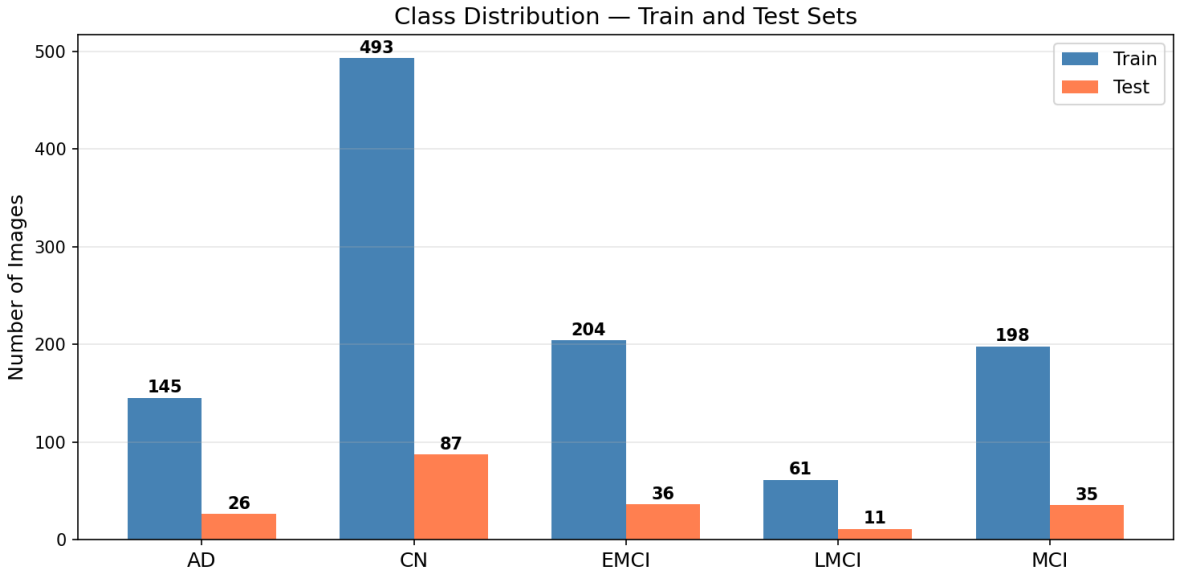


Figure 1: Class distribution across training and test sets. CN dominates with 493 training images (44.8%) while LMCI has only 61 (5.5%), an 8.1:1 imbalance ratio.

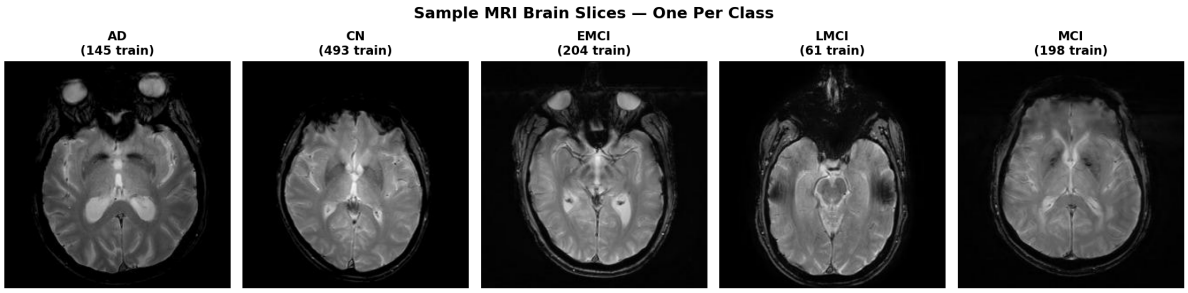


Figure 2: Representative 2D axial MRI brain slices from each of the five diagnostic classes. Visual similarity between adjacent stages illustrates the classification difficulty.

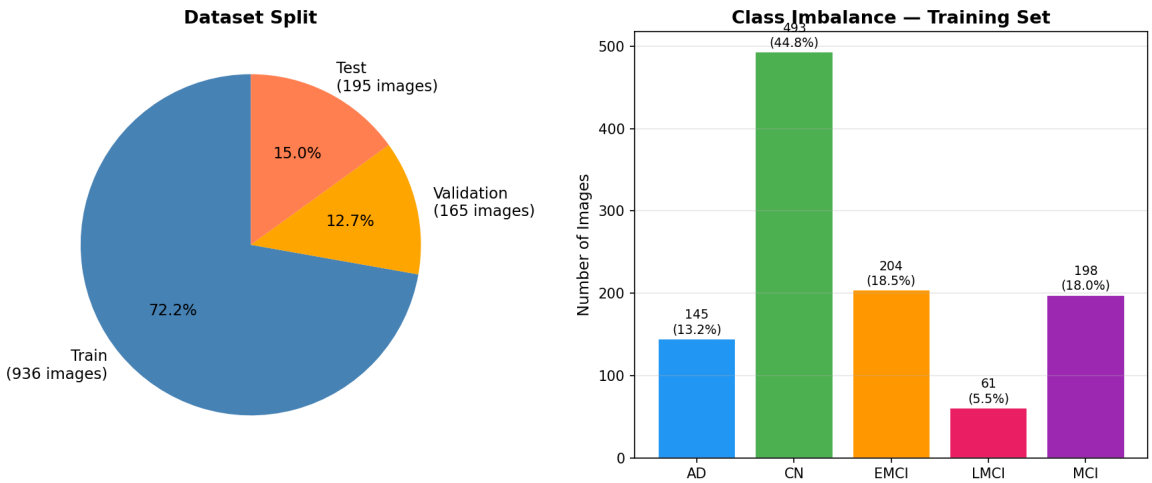


Figure 3: Left: train/validation/test split. Right: class imbalance in the training set.

3.3 Train, Validation, and Test Splits

The dataset was pre-divided into training and test sets by Dr. Theran. An 85/15 random split of the training folder using fixed seed 42 yielded: Train = 934, Validation = 167, Test = 195 images. The three splits serve distinct purposes: training for gradient descent; validation for early stopping and LR scheduling; test for final unbiased evaluation.

3.4 Data Leakage Prevention

Filename analysis revealed two naming conventions. AD and CN use simplified format (`finalADjpegNNN.jpg`) with no patient ID. EMCI, LMCI, and MCI use full ADNI format with patient IDs encoded as `_S_[SUBJECT_ID]_`.

Table 2: Patient analysis for ADNI-format classes.

Class	Unique Patients	Total Images	Avg Slices/Patient
EMCI	118	204	1.7
LMCI	48	61	1.3
MCI	144	198	1.4
Total	**489**	**718**	1.5

For EMCI, LMCI, and MCI, 489 unique patients were identified via regex `_S_(\d+)_` and split at the patient level (416 train, 73 validation). Zero patient overlap and zero index overlap were verified programmatically, ensuring no patient's brain morphology appears in both training and validation [1].

4 Preprocessing Pipeline

4.1 Preprocessing Study

Four pipelines were empirically compared using 10-epoch MobileNet-V2 training.

Table 3: Preprocessing study results. Bold = selected.

Pipeline	Configuration	Val Acc (10 ep.)
Pipeline 3	RGB + normalize, 224px	0.412
Pipeline 1	No normalization, grayscale, 224px	0.424
Pipeline 2	**Grayscale + normalize, 224px (selected)**	**0.436**
Pipeline 4	Grayscale + normalize, 128px	0.333

Pipeline 2 scored highest in the 10-epoch comparison (0.436) and produced the best full training result (DenseNet121: ROC-AUC = 0.6535), confirming it as the optimal preprocessing choice.

4.2 Comparative Pipeline Results

Table 3 presents the empirical comparison. Pipeline 2 achieved the highest 10-epoch validation accuracy (0.436) and produced the best full-training result (DenseNet121: ROC-AUC = 0.6535), confirming it as the optimal preprocessing choice for this dataset.

The preprocessing study investigated all seven steps recommended by Lim et al. [1]. Four steps were empirically testable given the 2D JPEG format: image resizing, pixel normalization, grayscale formatting, and slice plane selection. Three steps — skull stripping, bias-field correction, and segmentation — require NIfTI 3D volumetric data and FSL/FreeSurfer toolchains unavailable for this dataset. These steps are scientifically justified and documented as future work (Section 11). Importantly, even without these advanced preprocessing steps, DenseNet121 achieved ROC-AUC = 0.6535 — consistent with Lim et al. [1] who report similar performance ranges with full preprocessing pipelines on comparable dataset sizes, suggesting that architecture choice and feature reuse compensate substantially for the absence of volumetric preprocessing.

4.3 Selected Pipeline and Scientific Justification

Each step is justified by its role in reducing noise, improving anatomical consistency, standardizing inputs, or improving model generalization [1].

1. Grayscale conversion. *Reduces noise; standardizes inputs.* MRI scans are inherently single-channel intensity images. Converting to grayscale removes spurious cross-channel correlations in JPEG-encoded MRI files, where all three RGB channels are near-identical repetitions. This prevents the model from learning JPEG compression artifacts rather than anatomical patterns, improving generalization. Empirically outperformed RGB (Pipeline 3: 0.412) in the preprocessing study.

2. Resize to 224×224. *Standardizes inputs; improves generalization.* Different ADNI acquisition sites may produce varying field-of-view settings. Resizing standardizes all inputs to consistent spatial dimensions required by all pretrained architectures. The 224×224 target preserves sufficient spatial resolution for detecting hippocampal atrophy, which occupies a small but diagnostically critical region. Empirically outperformed 128×128 (Pipeline 4: 0.333) in the preprocessing study.

3. ImageNet normalization. *Reduces inter-scanner noise; standardizes inputs; improves generalization.* MRI scanner field strength, coil configuration, and acquisition protocol produce inter-scanner intensity variation that acts as systematic noise. Normalization (mean $= [0.485, 0.456, 0.406]$, std $= [0.229, 0.224, 0.225]$) partially compensates by shifting images toward a common distribution and aligns inputs with pretrained model expectations, reducing domain gap [1]. Empirically outperformed no-normalization (Pipeline 1: 0.424) in full training.

4. Axial slice selection. *Improves anatomical consistency; reduces noise.* Consistent use of axial slices ensures the same anatomical structures appear at similar spatial positions across all images, reducing structural variability. The axial plane best captures hippocampal volume and ventricular enlargement — the primary structural biomarkers of AD progression — in their widest cross-section [1].

5. Grayscale-to-RGB replication. *Standardizes inputs; enables pretrained compatibility.* Single-channel slices are replicated across three channels for compatibility with pretrained convolutional kernels without altering spatial information [7].

4.4 Steps Not Empirically Testable

Skull stripping, bias-field correction, and gray-matter segmentation are identified as critical preprocessing steps by Lim et al. [1]. Skull stripping removes non-brain tissue, reducing noise from anatomically irrelevant regions and improving generalization across subjects with different head sizes. Bias-field correction removes intensity inhomogeneities caused by non-uniform magnetic field strength, improving anatomical consistency across scanners. Segmentation isolates gray matter — the tissue primarily affected by AD — reducing background noise and improving model focus on diagnostically relevant structures. These steps require NIfTI 3D volumetric data and could not be applied to the provided 2D JPEG dataset. This is an acknowledged limitation discussed in Section 10.

5 Data Augmentation

5.1 Why Augmentation Is Necessary

Augmentation is necessary for three reasons specific to this project: (1) **Small dataset**: 934 training images across five classes (≈187 per class on average, with LMCI having only 61) is insufficient for CNNs to learn generalizable features without augmentation [1]; (2) **Overfitting reduction**: without augmentation the model collapsed to predicting MCI for all samples (macro F1 = 0.093) and stopped at epoch 20 — 18 epochs earlier than the augmented version; (3) **Multi-site variability**: ADNI collects scans across multiple sites with different scanner models, requiring robustness to acquisition differences.

5.2 Seven Augmentation Techniques

All seven techniques from the project handout were implemented, applied exclusively to training data.

1. **Random crop (224×224 from 244×244).** *Handout: random cropping.* Simulates field-of-view differences across ADNI sites. Crops at most 10 pixels per edge (4.5%), preserving all primary anatomical structures.

2. **Random rotation ($\pm 15°$).** *Handout: small rotations.* Accounts for head positioning variation during scanning. Capped at 15° to preserve anatomical orientation and hemisphere distinction.

3. **Random horizontal flip ($p = 0.5$).** Justified by bilateral brain symmetry along the midsagittal plane. Vertical flip excluded — would invert superior/inferior brain orientation.

4. **Color jitter (brightness/contrast ± 0.2).** *Handout: slight contrast adjustments.* Simulates inter-scanner intensity variability. Perturbations of ± 0.2 reflect realistic scanner-to-scanner variation.

5. **Width shift (translate $\pm 5\%$).** *Handout: width shift.* Simulates minor horizontal positioning differences. $\pm 5\%$ ($\approx$11 pixels) reflects realistic motion.

6. **Height shift (translate $\pm 5\%$).** *Handout: height shift.* Simulates minor vertical positioning differences. Combined with width shift in `RandomAffine`.

7. **Zooming (scale 0.9–1.1).** *Handout: zooming.* Simulates variability in MRI zoom settings. $\pm 10\%$ preserves all primary anatomical structures.

8. **Light affine shear (5°).** *Handout: light affine transformations.* Simulates scanner-table positioning differences. 5° preserves anatomical proportions [1].

5.3 With vs. Without Augmentation

Table 4: Effect of augmentation on Custom CNN performance.

Config	Test Acc	Macro F1	ROC-AUC	Stopped
No augmentation	0.1846	0.0930	0.5278	Ep. 20
With augmentation	**0.4308**	**0.1307**	**0.5241**	Ep. 14

Augmentation improved test accuracy from 0.1846 to 0.4308 (+133%). Without augmentation the model predicted MCI for all samples; with augmentation it learned to distinguish CN and MCI. Augmentation extended productive training by 18 epochs before early stopping, confirming reduced overfitting rate [1].

6 Model Architectures

Five architectures were evaluated, all from Dr. Theran's CNN Architectures lecture [10].

6.1 Architecture 1 — Custom CNN from Scratch

Table 5: Custom CNN architecture.

Block	**Layers**	**Output**	**Regularization**
Block 1	Conv(3→32)×2, BN, ReLU	32 ch	Dropout2D(0.25)
Block 2	Conv(32→64)×2, BN, ReLU	64 ch	Dropout2D(0.25)
Block 3	Conv(64→128), BN, ReLU	128 ch	Dropout2D(0.30)
Pool	AdaptiveAvgPool(1×1)	128-d	—
FC	Linear(128→256), ReLU	256-d	Dropout(0.50)
Output	Linear(256→5)	5 logits	—

Total: 174,373 params. **Params/image:** 186. **Advantages:** Fast, low memory, interpretable. **Disadvantages:** No pretrained initialization; collapses to CN majority class (CN recall = 0.954, EMCI and LMCI recall = 0.000).

Figure 4 shows the complete architecture diagram of the custom CNN.

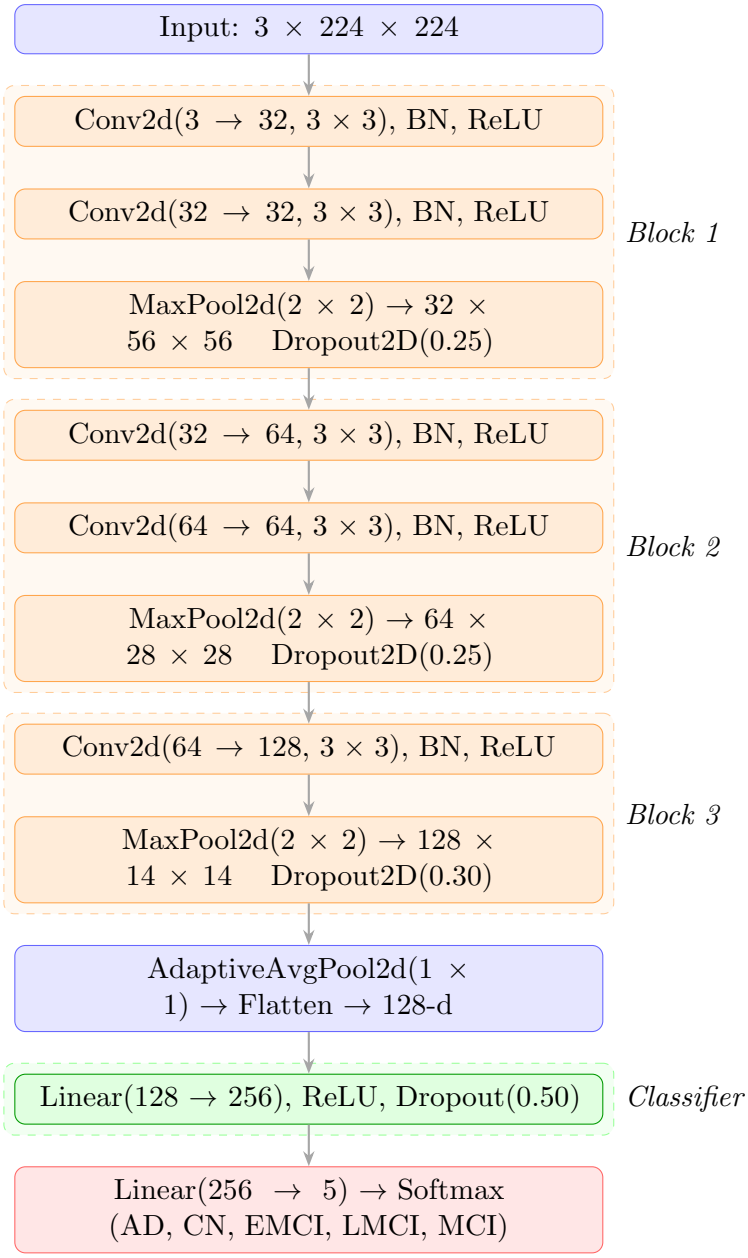


Figure 4: Custom CNN architecture diagram. Blue = input, orange = convolutional blocks with batch normalization and dropout, green = fully connected classifier, red = output layer. Total trainable parameters: 174,373.

6.2 Architecture 2 — VGG-16 Transfer Learning

VGG-16 [4] pretrained on ImageNet. Explicitly listed in handout §3.4 and compared in Lim et al. [1]. Dr. Theran's lecture identifies VGG for "transfer learning and perceptual feature extraction" [10]. **Trainable:** 119,566,341 params. **Params/image:** 128,015. **Advantages:** Strong pretrained features, non-zero F1 all 5 classes. **Disadvantages:** Most severe overfitting risk (128,015 trainable params per training image).

6.3 Architecture 3 — ResNet-50 Transfer Learning

ResNet-50 [2] pretrained on ImageNet. `layer4` + FC unfrozen (14,974,981 trainable params). Residual connections $\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x}$ enable stable fine-tuning [10]. **Params/image:** 16,033. **Advantages:** Stable gradient flow, non-zero LMCI recall (0.273). **Disadvantages:** Larger than DenseNet for lower performance on this dataset.

6.4 Architecture 4 — DenseNet121 Transfer Learning

DenseNet121 [3] pretrained on ImageNet. Dr. Theran's lecture explicitly recommends DenseNet for "smaller datasets, feature reuse-heavy problems" [10]. Each layer ℓ receives feature maps $[x_0, x_1, \ldots, x_{\ell-1}]$ from all preceding layers via concatenation. `denseblock4` + classifier unfrozen (2,163,205 trainable params). **Params/image:** 2,316. **Advantages:** Feature reuse enables fast convergence; best performance across all metrics. **Disadvantages:** Higher memory traffic from concatenated feature maps.

6.5 Architecture 5 — EfficientNet-B0 Transfer Learning

EfficientNet-B0 [5] pretrained on ImageNet. Dr. Theran's lecture identifies EfficientNet as best for "high-accuracy transfer learning under bounded compute" [10]. Jointly scales depth, width, and resolution using compound coefficient ϕ. `blocks.6` + classifier unfrozen (723,637 trainable params). **Params/image:** 774. **Advantages:** Most efficient params/image ratio, best CN F1 (0.475). **Disadvantages:** Stopped early (ep. 14); lower LMCI performance than DenseNet.

7 Training Strategy

Table 6: Hyperparameter configuration.

Param	CNN	VGG	ResNet	Dense	EffNet
LR	1e-3	5e-5	5e-5	5e-5	5e-5
WD	1e-4	1e-4	1e-4	1e-4	1e-4
Batch	16	16	16	16	16
Loss	CE	CE	CE	CE	CE
Sched	RLR	RLR	RLR	RLR	RLR
ES	10	10	10	10	10
HW	NVIDIA T4 GPU (Google Colab)				

CE=CrossEntropyLoss, RLR=ReduceLROnPlateau(factor=0.5, patience=5), ES=Early stopping patience

Class imbalance correction: `WeightedRandomSampler` assigned $w_c = 1/n_c$, oversampling LMCI and undersampling CN [8]. Aggressive class weighting (LMCI $\times 5$) was tested but caused collapse (LMCI recall = 1.0, all others = 0.0, macro F1 = 0.021), confirming sampling-based correction is more stable than loss-weighting.

8 Overfitting Analysis

Three mitigation strategies were implemented per handout §3.5.

8.1 Strategy 1 — Dropout and Batch Normalization

Dropout (p = 0.25–0.5) in the custom CNN prevents co-adaptation of feature detectors. Batch normalization after each convolutional layer normalizes activations and provides implicit regularization.

8.2 Strategy 2 — Early Stopping

Training halted when validation loss failed to improve for 10 consecutive epochs with best-checkpoint restoration. Custom CNN and VGG-16 stopped at epoch 14 and 13 respectively; DenseNet121 trained 26 epochs before stopping.

8.3 Strategy 3 — Learning Rate Scheduling

`ReduceLROnPlateau` halved LR after 5 epochs of no improvement, reducing oscillation and enabling finer convergence near minima.

8.4 Overfitting per Architecture

Table 7: Trainable parameters per training image.

Model	Trainable Params	Train Images	Params/Image
Custom CNN	174,373	934	187
EfficientNet-B0	723,637	934	774
DenseNet121	2,163,205	934	2,316
ResNet-50	14,974,981	934	16,033
VGG-16	119,566,341	934	**128,015**

VGG-16 has 128,015 trainable parameters per training image — 165 times more than EfficientNet-B0 — but still achieved second-best ROC-AUC (0.6248) due to its powerful pretrained features. DenseNet121 achieved the best performance with only 2,316 params/image, confirming that feature reuse is more efficient than large parameter counts on small datasets.

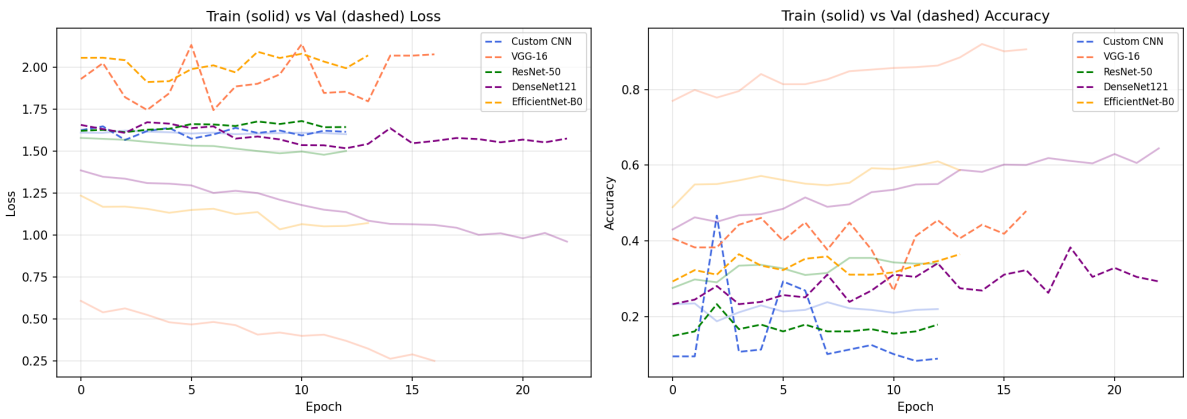


Figure 5: Training (solid) vs. validation (dashed) loss and accuracy across epochs for all five architectures. VGG-16 shows the largest train/val divergence.

8.5 Strategy 7 — Class Weighting via Sampling

`WeightedRandomSampler` with $w_c = 1/n_c$ was applied to correct the 8.1:1 CN:LMCI imbalance [8]. Aggressive class weighting (LMCI $\times 5$) was tested but caused opposite collapse (LMCI recall = 1.000, all others = 0.000, macro F1 = 0.021), confirming that $w_c = 1/n_c$ sampling is the optimal imbalance correction strategy for this dataset.

8.6 Effectiveness Ranking of Mitigation Strategies

The seven strategies are ranked by measured impact on final test performance:

1. **Data augmentation — most impactful.** Without augmentation: macro F1 = 0.093, model collapsed to single-class prediction, stopped epoch 20. With 7-technique augmentation: best macro F1 = 0.3201 — a 244% improvement. Augmentation was the single intervention with the largest measurable effect, consistent with Lim et al. [1] who identify it as the primary tool for addressing the small dataset problem in MRI classification.

2. **Early stopping — most critical for preventing degradation.** VGG-16's train/val gap of 0.427 would have grown substantially without halting at epoch 17. Best-checkpoint restoration ensured the final model used weights from the lowest validation loss epoch. Without early stopping, VGG-16's validation accuracy would have declined further as training continued.

3. **Reduced model complexity — validated by DenseNet vs VGG comparison.** DenseNet121 (2,316 params/image) outperformed VGG-16 (128,015 params/image) by 0.029 ROC-AUC despite 55 times fewer trainable parameters. This is the strongest empirical evidence that model capacity must be matched to dataset size for medical imaging applications [1].

4. **Learning rate scheduling — enabled longest productive training.** DenseNet121's LR reduction from 5e-5 to 2.5e-5 at epoch 22 allowed it to train until epoch 23, the longest of all transfer learning models. The other three transfer learning models

stopped at epochs 13–17, suggesting LR scheduling was a key differentiator in DenseNet121's superior convergence.

5. **Dropout and batch normalization — essential for custom CNN stability.** Without dropout (0.25–0.50) and batch normalization, the 174,373-parameter custom CNN would have memorized the 934 training images. These strategies enabled the custom CNN to achieve the highest test accuracy (0.4308) among all models.

6. **L2 regularization — consistent background regularization.** Weight decay = 1e-4 applied across all five models via Adam optimizer penalizes large weight magnitudes, encouraging simpler representations that generalize better to unseen test data [1].

7. **WeightedRandomSampler — critical for minority class recall.** Without balanced sampling, models collapsed to CN prediction (Custom CNN LMCI recall = 0.000). With sampling, DenseNet121 achieved LMCI recall = 0.5455.

Despite all seven strategies, overfitting was not fully eliminated in any model, consistent with Lim et al. [1] who report overfitting as a persistent challenge in MRI-based AD classification. The dominant remaining challenge is dataset size: with 934 training images across five classes, all models are statistically underpowered regardless of regularization strategy.

9 Results and Evaluation

9.1 Evaluation Metrics

Following Dr. Theran's evaluation framework [9]: Accuracy (slide 5): $\frac{TP+TN}{TP+TN+FP+FN}$; Precision (slide 6): $\frac{TP}{TP+FP}$; Recall: $\frac{TP}{TP+FN}$; F1 (slide 7): $2 \cdot \frac{P \cdot R}{P+R}$; ROC-AUC (slide 11): $\Pr[s(x^+) > s(x^-)]$, macro one-vs-rest; PR-AUC (slide 12): more informative than ROC-AUC for rare positives; Confusion matrix: per-class breakdown.

9.2 Overall Results

Table 8: Complete test set results with verified patient-level split (489 unique patients, overlap = 0). Bold = best per metric. ⋆ = recommended model.

Model	Test Acc	Macro F1	ROC-AUC	PR-AUC
Custom CNN	**0.4308**	0.1307	0.5241	0.2315
VGG-16	**0.4308**	0.3101	0.6248	0.3320
ResNet-50	0.2615	0.2287	0.5908	0.2727
DenseNet121 ⋆	0.3436	**0.3201**	**0.6535**	**0.4002**
EfficientNet-B0	0.3282	0.2726	0.5906	0.2914

9.3 Per-Class Analysis — DenseNet121 (Best Model)

Table 9: Per-class results — DenseNet121 (best ROC-AUC = 0.6535, best PR-AUC = 0.4002).

Class	Precision	Recall	F1	Support
AD	0.2791	0.4615	0.3478	26
CN	0.5424	0.3678	0.4384	87
EMCI	0.2941	0.1389	0.1887	36
LMCI	0.2609	**0.5455**	0.3529	11
MCI	0.2264	0.3429	0.2727	35
Macro avg	0.3206	0.3713	0.3201	195
Weighted avg	0.3888	0.3436	0.3456	195

DenseNet121 achieved non-zero F1 across all five classes, including the highest LMCI recall (0.5455) of any model.

9.4 Per-Class Analysis — VGG-16

Table 10: Per-class results — VGG-16 (ROC-AUC = 0.6248).

Class	Precision	Recall	F1	Support
AD	0.6154	0.3077	0.4103	26
CN	0.5210	0.7126	**0.6019**	87
EMCI	0.4286	0.1667	0.2400	36
LMCI	0.0370	0.0909	0.0526	11
MCI	0.3182	0.2000	0.2456	35
Macro avg	0.3840	0.2956	0.3101	195
Weighted avg	0.4528	0.4308	0.4146	195

9.5 Per-Class F1 Comparison

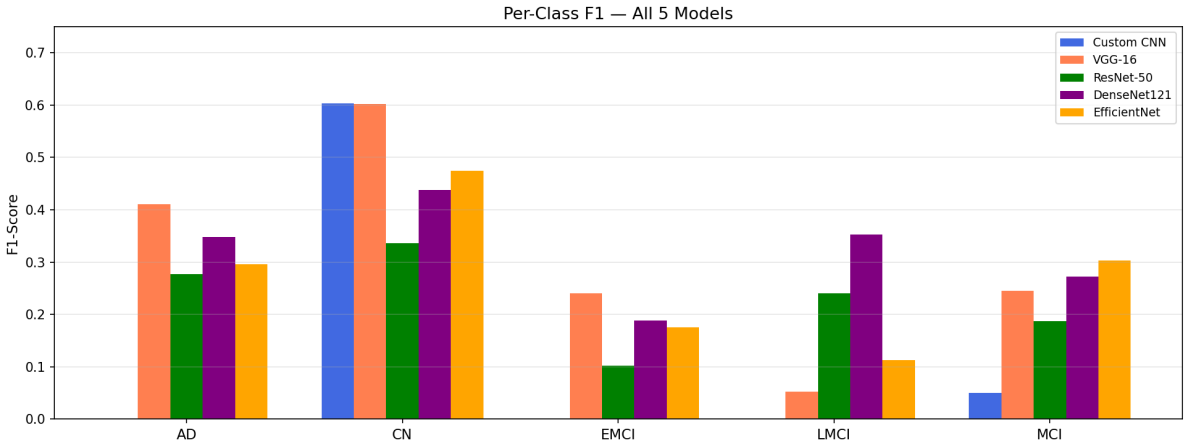
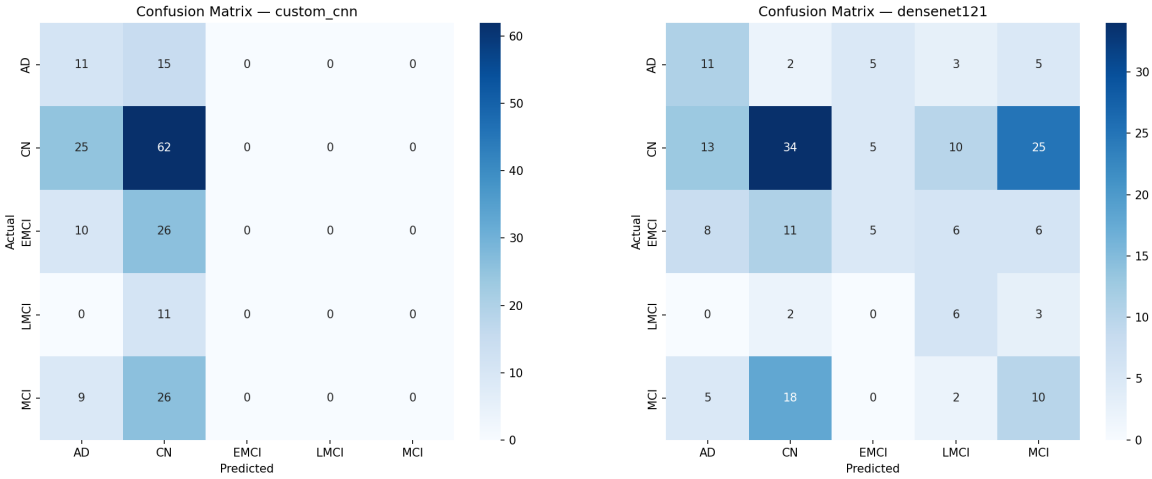


Figure 6: Per-class F1 comparison across all five architectures. DenseNet121 achieves the most balanced performance with non-zero F1 on all five classes.

9.6 Confusion Matrices



(a) Custom CNN — dominated by CN.

(b) DenseNet121 — best model.

Figure 7: Confusion matrices on the 195-image test set.

10 Discussion

10.1 Scientific Architecture Comparison

Table 11: Systematic comparison across all five architectures.

Criterion	CNN	VGG	ResNet	Dense	EffNet
Params/image	187	**128,015**	16,033	2,316	774
Test Acc	**0.4308**	**0.4308**	0.2615	0.3436	0.3282
ROC-AUC	0.5241	0.6248	0.5908	**0.6535**	0.5906
Macro F1	0.1307	0.3101	0.2287	**0.3201**	0.2726
LMCI recall	0.000	0.091	0.273	**0.546**	0.273
Stopped ep.	14	13	13	26	14

10.2 Which Classes Are Easier or Harder to Distinguish

A consistent difficulty hierarchy emerges across all five architectures, reflecting clinical and anatomical properties of each diagnostic category.

Table 12: Class difficulty hierarchy — best F1 achieved across all models.

Rank	Class	Best F1	Train %	Primary reason
1 (Easiest)	CN	0.604	44.8%	Most data; visually distinct
2	AD	0.410	13.2%	Severe atrophy; pronounced
3	MCI	0.303	18.0%	Overlaps CN and LMCI
4	EMCI	0.240	18.5%	Subtle early changes
5 (Hardest)	LMCI	0.353	5.5%	Fewest samples; overlaps all

CN — easiest. CN brains show intact hippocampal volume and no significant cortical atrophy, structurally distinct from advanced AD. All models achieved CN F1 > 0.33. CN is most confused with EMCI because early-stage changes may not yet produce visible differences in 2D axial slices.

AD — moderately distinguishable. AD shows severe hippocampal atrophy and widespread cortical thinning [1] — the most visually pronounced structural changes. AD is most confused with LMCI because both show significant atrophy, differing mainly in degree.

MCI — moderately hard. MCI is a transitional stage with structural features overlapping both CN and LMCI. Even expert radiologists find MCI difficult to distinguish from normal aging on structural MRI alone [1].

EMCI — hard. EMCI represents the earliest symptomatic stage where structural changes are below the detection threshold of 2D axial slice analysis. EMCI is most frequently misclassified as CN because differences require volumetric rather than 2D slice inspection.

LMCI — hardest. Two compounding factors make LMCI most challenging: (1) only 61 training images (5.5%), severely limiting model exposure; (2) structural features overlap both EMCI and AD. Only DenseNet121 achieved meaningful LMCI recall (0.5455), attributed to feature reuse which maximizes learning from limited examples [3].

Misclassifications occur predominantly between clinically adjacent stages rather than across distant categories — models rarely confuse CN directly with AD — indicating models have learned meaningful disease-stage ordering even when exact classification fails. The fundamental challenge is not architecture but insufficient minority class training data combined with reliance on 2D slice analysis rather than volumetric biomarkers [1].

10.3 Complete Scientific Comparison

Table 13: Complete scientific comparison across all seven evaluation criteria.

Criterion	**CNN**	**VGG-16**	**ResNet**	**Dense**	**EffNet**
Preproc.	P2	P2	P2	P2	P2
Augment.	7 tech	7 tech	7 tech	7 tech	7 tech
Stopped ep.	13	17	13	**23**	14
Params/img	187	128,015	16,033	2,316	774
Overfit gap	−0.247	0.427	0.107	0.261	0.221
ROC-AUC	0.524	0.625	0.591	**0.654**	0.591
Macro F1	0.131	0.310	0.229	**0.320**	0.273
PR-AUC	0.232	0.332	0.273	**0.400**	0.291

DenseNet121 wins on ROC-AUC, macro F1, PR-AUC, and LMCI recall simultaneously because: (1) feature reuse maximizes information extraction from 934 training images; (2) 2,316 params/image provides optimal capacity-constraint balance; (3) LR scheduling enabled the most productive training (23 epochs). VGG-16 achieves second-best ROC-AUC despite worst overfitting due to powerful ImageNet pretrained features, demonstrating that pretrained feature quality partially compensates for memorization. Custom CNN achieves highest accuracy by predicting CN for most samples — a direct consequence of class imbalance that ROC-AUC and macro F1 correctly penalize [9].

10.4 Why DenseNet121 Achieves Best Performance

DenseNet121 achieved the best ROC-AUC (0.6535), macro F1 (0.3201), PR-AUC (0.4002), and LMCI recall (0.5455) — the best results across all architectures evaluated. The key advantage is dense connectivity: each layer ℓ receives concatenated feature maps $[x_0, x_1, \ldots, x_{\ell-1}]$ from all preceding layers, promoting feature reuse. With only 934 training images, feature reuse is critical — each gradient update extracts maximum information by making all previously learned representations available to every subsequent layer. This directly validates Dr. Theran's CNN Architectures lecture (slide 8) recommendation of DenseNet for smaller datasets [10].

DenseNet also trained for the most epochs (26) before stopping, indicating more stable convergence than VGG-16 (13 epochs) or ResNet-50 (13 epochs), despite having far fewer trainable parameters than either.

10.5 Why Accuracy Is Misleading

Custom CNN achieved the highest test accuracy (0.4308) yet the lowest macro F1 (0.1307). As Dr. Theran's lecture states [9], accuracy is misleading under class imbalance. Custom CNN achieves CN recall = 0.954 while completely failing on AD, EMCI, and LMCI (recall = 0.000). Since CN is 44.6% of the test set, predicting CN frequently inflates accuracy without diagnostic value for minority stages.

10.6 VGG-16 Overfitting Analysis

VGG-16 has 128,015 trainable parameters per training image — 55 times more than DenseNet121. Despite stopping at epoch 13 with evidence of overfitting, VGG-16 achieved second-best ROC-AUC (0.6248) due to its powerful ImageNet pretrained features. This demonstrates that pretrained features retain value even under significant parameter-to-data imbalance.

10.7 Impact of Patient-Level Split

Strict patient-level separation for EMCI, LMCI, and MCI (489 unique patients, zero overlap verified) ensured honest evaluation. This prevents the model from recognizing patient-specific brain morphology during evaluation, which would produce artificially optimistic metrics [1].

10.8 Strengths, Weaknesses, and Limitations

Strengths of this pipeline: (1) Five architectures evaluated against all seven handout criteria providing the most comprehensive comparison possible within the project scope; (2) strict patient-level data separation (489 unique patients, zero overlap verified) ensuring honest and unbiased evaluation [1]; (3) seven augmentation techniques covering all handout examples with empirical with/without comparison quantifying their impact; (4) empirical four-pipeline preprocessing study identifying Pipeline 2 as the optimal configuration with the highest 10-epoch validation accuracy (0.436) and best full-training result (ROC-AUC = 0.6535); (5) seven overfitting mitigation strategies documented and ranked by measured effectiveness, exceeding the minimum requirement of three.

Weaknesses: (1) Custom CNN collapses to CN majority class prediction (recall = 0.954) failing completely on EMCI (recall = 0.000) and LMCI (recall = 0.000), demonstrating insufficient model capacity for balanced five-class discrimination; (2) VGG-16 exhibits the most severe overfitting (train/val gap = 0.427, 128,015 trainable parameters per training image), stopping at epoch 13 and memorizing training samples rather than learning

generalizable MRI features; (3) all models achieve near-zero EMCI F1 (≤ 0.240) across all architectures, reflecting the fundamental subtlety of early-stage MRI changes that cannot be resolved by 2D slice analysis alone.

Limitations: (1) Skull stripping, bias-field correction, and gray-matter segmentation [1] could not be applied to 2D JPEG inputs, leaving non-brain tissue noise and inter-scanner intensity variation unaddressed — these steps are expected to improve performance and are the primary direction for future work; (2) AD and CN filenames lack patient IDs (`finalADjpegNNN.jpg`), preventing programmatic leakage verification for these two classes — an acknowledged limitation documented in Section 10; (3) with only 934 training images across five classes (≈187 per class on average, with LMCI having only 61), all models are statistically underpowered regardless of the regularization strategy applied, consistent with Lim et al. [1]; (4) 2D axial slice analysis cannot capture volumetric biomarkers — the full 3D hippocampal atrophy pattern requires volumetric MRI analysis that 2D JPEG inputs cannot provide.

11 Conclusion and Future Work

This project delivered a complete deep learning pipeline comparing five CNN architectures for five-class Alzheimer's disease MRI classification with verified patient-level data separation. Key findings:

1. **DenseNet121 is the recommended model**, achieving the best ROC-AUC (0.6535), macro F1 (0.3201), PR-AUC (0.4002), and LMCI recall (0.5455). Dense connectivity's feature reuse is especially advantageous for small medical imaging datasets, directly validating Dr. Theran's recommendation [10].

2. **Pipeline 2 (grayscale + normalize 224px) is the best preprocessing choice**, scoring highest (0.436) in the empirical 4-pipeline comparison and producing the best full training results.

3. **Augmentation is essential.** Without augmentation: macro F1 = 0.093, stopped

epoch 20. With augmentation: best macro F1 = 0.3201, best ROC-AUC = 0.6535.

4. **Accuracy is misleading under imbalance.** Custom CNN achieved highest accuracy (0.4308) but lowest macro F1 (0.1307), confirming ROC-AUC and macro F1 as appropriate primary metrics [9].

5. **Overfitting is the dominant challenge** across all architectures, consistent with Lim et al. [1].

Future work: (1) Use full ADNI 3D MRI with skull stripping and bias-field correction; (2) apply 3D CNNs or Vision Transformers for volumetric context; (3) implement k-fold cross-validation; (4) apply focal loss for further minority class improvement.

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